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# Omega Theorems for Zeta Sums

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**Abstract**—We obtain lower bounds for the moduli of trigonometric sums in the theory of Riemann zeta functions.

KEY WORDS: *Riemann zeta function, trigonometric sum, zeta sum, omega theorem, Vinogradov method.*

We shall use the usual notation; in particular, $\zeta(s)$ is the Riemann zeta function, $s = \sigma + it$, $i^2 = -1$, $\sigma, t$ are real numbers; $c, c_1, \dots$ are absolute positive constants, in general, different in different assertions, the constants in the signs $\ll$ are absolute.

By *zeta sums* we mean trigonometric sums $S$ of the form (see [1, p. 66])

$$S = \sum_{n \le N} n^{it}, \qquad S = \sum_{a < n \le b} n^{it}.$$

Here $t$ is an increasing parameter, $t \ge t_1 > 0$; $N = N(t) \le t$, or, often, $N \le \sqrt{t}$; in the same way, $0 < a < b \le t$ or $b \le \sqrt{t}$. The number of summands in $S$ is called the *length of the sum*, and so the greater the number of summands, the "longer" it is. The sums $S$ and the upper bounds for $|S|$ are closely related to the order of growth of $|\zeta(\sigma + it)|$, $t \to +\infty$, where $\sigma$ is a fixed number (see [2, p. 112]). It is natural to pose the question of uniform (in $t$ and $N$) estimates of zeta sums and state omega theorems for them. Similar questions, but only for complete rational trigonometric sums, were studied by the author in [3]. It should also be noted that in [4] the author proved the omega theorem for the remainder in the multidimensional Dirichlet divisor problem; moreover, the form of the remainder is the form obtained by Dirichlet, i.e., it is uniform in the number of summands in the sum and in the number of factors defining the divisor function. This omega theorem also involves zeta sums.

Averaging (over $t$) the square of the modulus of $S$ shows that the "bulk" of $|S|$ can be estimated from above slightly less accurately than the quadratic root of the number of summands. Indeed, if $T \ge T_1 > 0$, $2 \le N \le \sqrt{T}$ and the set $E \subset (T, 2T)$ is the set of the numbers $t$ for which the inequality

$$\left|\sum_{n \le N} n^{it}\right| \ge \sqrt{N}\, h(T), \qquad h(T) > 0,$$

holds, then we can easily find that

$$Nh^2(T)\mu(E) \le \int_E \left|\sum_{n \le N} n^{it}\right|^2 dt \le \int_T^{2T} \left|\sum_{n \le N} n^{it}\right|^2 dt \le TN + 16N \log N \le 17TN.$$

Hence, for the measure $\mu(E)$ we obtain the estimate

$$\mu(E) \le T\Delta, \qquad \Delta = 17h^{-2}(T).$$

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If we take $h(T) = \log^c T$, $c > 0$, then we see that for all $t \in (T, 2T)$, with the exception of $t \in E \subset (T, 2T)$, $\mu(E) \le 17T \log^{-2c} T$, the following estimate holds:

$$\Bigl|\sum_{n\le N} n^{it}\Bigr| \le \sqrt{N}\,\log^c T. \tag{1}$$

It will be shown later that there exists an infinite sequence of numbers $T$, $T \to +\infty$, and an infinite sequence of numbers $N \le \sqrt{T}$, $N \to +\infty$, such that the lower bound of the corresponding zeta sums is much greater than the right-hand side of (1). To prove Theorem 1, we need a lower bound for $|\zeta(1/2+it)|$. Let us state only a special case of a more general assertion (see [5, p. 46]) as Lemma 1.

**Lemma 1.** *There exists a constant $T_0 > 0$ such that for $T \ge T_0$ the following inequality holds*:

$$\max_{T\le t\le 2T}\Bigl|\zeta\Bigl(\frac{1}{2}+it\Bigr)\Bigr| \ge \exp\biggl(\frac{3}{4}\sqrt{\frac{\log T}{\log\log T}}\biggr).$$

**Theorem 1.** *There exists an infinite sequence of numbers $t_\nu$, $\nu = 1, 2, \dots$, satisfying the inequalities $t_1 \ge 1$, $t_{\nu+1}t_\nu^{-1} \ge 2$ and possessing the following property*: *for each $t = t_\nu$, on the interval $(1, P)$, $P = [\sqrt{t/(2\pi)}\,]$, there exists at least $K$ positive integers $N$,*

$$K \ge f^2(t), \qquad \textit{where} \quad f(t) = \exp\biggl(\frac{3}{5}\sqrt{\frac{\log t}{\log\log t}}\biggr),$$

*such that the following inequality holds*:

$$\Bigl|\sum_{n\le N} n^{it}\Bigr| \ge \sqrt{N}\,f(t).$$

**Proof.** In Lemma 1, we take $T = 4^\nu T_0$, $\nu = 1, 2, \dots$. The sequence of numbers $t_\nu$, whose existence is asserted in the theorem, is defined by the relation

$$\max_{T\le t\le 2T}\Bigl|\zeta\Bigl(\frac{1}{2}+it\Bigr)\Bigr| = \Bigl|\zeta\Bigl(\frac{1}{2}+it_\nu\Bigr)\Bigr|.$$

Obviously, $t_{\nu+1}t_\nu^{-1} \ge 2$, $\nu = 1, 2, \dots$. Further, we set $t = t_\nu$. Suppose that, for each positive integer $N$, $1 \le N \le P$, the following inequality holds:

$$\Bigl|\sum_{n\le N} n^{it}\Bigr| \le 2\sqrt{N}\,f(t). \tag{2}$$

Using the Hardy–Littlewood approximate equation, we obtain

$$\Bigl|\zeta\Bigl(\frac{1}{2}+it\Bigr)\Bigr| \le 2\Bigl|\sum_{n\le P}\frac{1}{\sqrt{n}}\,n^{it}\Bigr| + O(t^{-1/4}).$$

Setting

$$S(n) = \sum_{m\le n} m^{it},$$

we find that

$$\sum_{n\le P}\frac{1}{\sqrt{n}}\,n^{it}=\frac{1}{\sqrt{P}}\,S(P)+\sum_{1<n\le P}\biggl(\frac{1}{\sqrt{n-1}}-\frac{1}{\sqrt{n}}\biggr)S(n-1).$$

Combining this with (2), we obtain the following estimates:

$$\sum_{n\le P}\frac{1}{\sqrt{n}}\,n^{it}\le 2f(t)+\sum_{1<n\le P}\frac{1}{n}2f(t)<2f(t)\,\log t,$$

$$\Bigl|\zeta\Bigl(\frac{1}{2}+it\Bigr)\Bigr|\le 5f(t)\,\log t,\qquad t\ge t_0>0.$$

But the last inequality contradicts the estimate in Lemma 1. Therefore, the assumption that for each $N$, $1\le N\le P$, inequality (2) holds is false: there exist numbers $N$ such that

$$\Bigl|\sum_{n\le N}n^{it}\Bigr|>2\sqrt{N}\,f(t).$$

Suppose that $N$ is one of them. From the trivial upper bound for $|S(N)|$, we obtain

$$N>2\sqrt{N}\,f(t),\qquad N>4f^2(t).$$

Further, for any integer $M\ge 0$ the following relation is valid:

$$S(N+M)=S(N)+\sum_{N<n\le N+M}n^{it}.$$

Passing to the inequalities, we have

$$|S(N+M)|\ge|S(N)|-M>2\sqrt{N}\,f(t)-M\ge\sqrt{N+M}\,f(t)$$

whenever $M\le\sqrt{N}\,f(t)/2$. Recalling that $N>4f^2(t)$, we find that there exists at least $K$, $K\ge f^2(t)$, numbers $n$ such that

$$|S(n)|\ge\sqrt{n}\,f(t).$$

Theorem 1 is proved. $\square$

It follows from Theorem 1 that, apparently, it is impossible to obtain a sharper estimate for the zeta sum than, for example, the following one:

$$|S(N)|\le\sqrt{N}\,\exp\Biggl(c\sqrt{\frac{\log t}{\log\log t}}\Biggr),\tag{3}$$

where $c>0$ is an absolute constant.

Upper bounds for $|S(N)|$ that are obtained even from the Riemann conjecture strongly differ from (3).

**Theorem 2.** *By the Riemann conjecture, we have the following estimate*:

$$|S(N)|\le\sqrt{N}\,\exp\biggl(c\frac{\log t}{\log\log t}\biggr),\tag{4}$$

*where $c>0$ is an absolute constant and $N\le t$.*

**Proof.** The proof of the theorem repeats the line of reasoning in the proof of Theorem 1 from [6, p. 153] (or in the solution of Problem 1 from [2, p. 112]) using the estimate of $|\zeta(1/2+it)|$, $t \ge t_1 > 0$ that follows from the Riemann conjecture (see for example, [7, p. 350 of the Russian translation]):

$$\left|\zeta\Big(\frac{1}{2}+it\Big)\right| \ll \exp\Big(A\frac{\log t}{\log\log t}\Big), \qquad A>0. \quad \square$$

The estimates (3) and (4) of the zeta sums are nontrivial whenever the inequalities

$$N \ge \exp\Big(c_1\sqrt{\frac{\log t}{\log\log t}}\Big), \qquad N \ge \exp\Big(c_1\frac{\log t}{\log\log t}\Big),$$

respectively, hold; here $c_1>0$ is an absolute constant and $t \ge t_1 > 0$. Recalling the following estimate of $|S(N)|$ obtained by Vinogradov's method (see [8] or [1, p. 95]):

$$|S(N)| \le N \exp\Big(-c\frac{\log^3 N}{\log^2 t}\Big), \tag{5}$$

we see that it is nontrivial whenever

$$N \ge \exp(c_1 \log^{2/3} t).$$

Therefore, Vinogradov's method allows us to obtain nontrivial estimate of shorter zeta sums than those estimated in Theorem 2 using the Riemann conjecture. But the estimates of short zeta sums allow us refine the boundary of zeros of $\zeta(s)$. Let us see what the estimate (3) yields in this case. To do this, we need the following Lemma 2 (see [7, p. 62 of the Russian translation]), which establishes the connection between the growth of $|\zeta(s)|$ and the boundary of zeros of $\zeta(s)$.

**Lemma 2.** *Suppose that, as* $t \to +\infty$,

$$\zeta(s) = O(e^{\varphi(t)})$$

*in the domain*

$$1-\theta(t) \le \sigma \le 2, \qquad t \ge t_1 > 0,$$

*where* $\varphi(t)$ *and* $\theta^{-1}(t)$ *are positive nondecreasing functions such that* $\theta(t) \le 1$ *and* $\varphi(t) \to +\infty$, *and, in addition,*

$$\frac{\varphi(t)}{\theta(t)} = o(e^{\varphi(t)}).$$

*Then there exists a constant* $A>0$ *such that* $\zeta(s)$ *has no zeros in the domain*

$$\operatorname{Re} s = \sigma \ge 1 - A\frac{\theta(2t+1)}{\varphi(2t+1)}, \qquad t \ge t_1 > 0.$$

**Theorem 3.** *When the estimate* (3) *is satisfied, the function* $\zeta(s)$ *has no zeros in the domain*

$$\operatorname{Re} s = \sigma \ge 1 - c_1\sqrt{\frac{\log\log t}{\log t}}, \qquad t \ge 14, \tag{6}$$

*where* $c_1 > 0$ *is an absolute constant.*

**Proof.** From (3) and the approximate functional equation for $\zeta(s)$, we obtain the following estimate for $|\zeta(\sigma+it)|$, $1/2 \le \sigma \le 1$, $t \ge 2$:

$$|\zeta(\sigma+it)| \ll \exp\biggl(c_2\sqrt{\frac{\log t}{\log\log t}}\biggr), \qquad c_2 > 0.$$

Setting

$$\varphi(t) = c_2\sqrt{\frac{\log t}{\log\log t}}, \qquad \theta(t) = \frac{1}{2},$$

in Lemma 2 we obtain the assertion of the theorem. □

Comparing (6) with the boundary of zeros of $\zeta(s)$ obtained by Vinogradov's method

$$\operatorname{Re} s = \sigma \ge 1 - c_1\frac{1}{(\log t)^{2/3}(\log\log t)^{1/3}}, \qquad t \ge 14,$$

we see that these two results do not differ much one from the other.

Another estimate of the zeta sums, whose special case was already examined above, is as follows:

$$|S(N)| \le N^{\beta} f(t), \tag{7}$$

where $\beta$ is a fixed number, $1/2 \le \beta < 1$. Using the Montgomery omega theorem (see [9]):

$$\zeta(\sigma+it) = \Omega\biggl(\exp\biggl(c\frac{(\log t)^{1-\sigma}}{(\log\log t)^{\sigma}}\biggr)\biggr), \qquad \frac{1}{2} < \sigma < 1,$$

and the arguments of Theorem 1, we conclude that, apparently, the sharpest estimate of the form (7) can be only one with

$$f(t) = \exp\biggl(c\frac{(\log t)^{1-\beta}}{(\log\log t)^{\beta}}\biggr). \tag{8}$$

Repeating the arguments of Theorem 3, we see that relations (7) and (8) yield the following boundary of zeros of $\zeta(s)$: $\zeta(s) \ne 0$ in a domain of the $s$-plane of the form

$$\operatorname{Re} s = \sigma \ge 1 - A\frac{(\log\log t)^{\beta}}{(\log t)^{1-\beta}}, \qquad t \ge 14,$$

where $A > 0$ is an absolute constant.

We can seek omega estimates of $|S(N)|$ of the form other than (7). Suppose that for $1 \le N \le t$ we have the estimate

$$|S(N)| \ll N\, \exp\biggl(-c\frac{\log^a N}{\log^b t}\biggr),$$

where $c > 0$ is a constant, $a > b > 0$, and $t \ge t_1 > 0$. For $a = 3$, $b = 2$ such an estimate was obtained by Vinogradov (see [8, 1]). Using the omega estimate of $|\zeta(1+it)|$ or the Dirichlet theorem, from the theory of Diophantine approximations we can easily prove that

$$\frac{a}{b} \ll \frac{\log\log t}{\log\log\log t}, \qquad t \to +\infty.$$

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